



Science of Corporate Technical Documentation for Software Engineers -Theories-

Composed By: Hayato Ishida

For Japanese version, click [here](#)

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About Me

- Accounts
 - GitHub: [@hayat01sh1da](#)
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 - LinkedIn: [@hayat01sh1da](#)
 - Speaker Deck: [@hayat01sh1da](#)
 - Docswell: [@hayat01sh1da](#)
- Occupation: Software Engineer
- Things I Am Into
 - Linguistics
 - Singing at Karaoke
 - Listening to Music
 - Watching Movies
 - Cat Feeding



Licences / Certifications

- English
 - TOEIC® Listening & Reading 915: Certified on December 2019
- Engineering
 - Information Security Management: Certified on November 2017
 - Applied Information Technology Engineer: Certified on June 2017
 - Fundamental Information Technology Engineer: Certified on November 2016
 - IT Passport: Certified on April 2016
- Others
 - Abacus 2nd Class: Certified on June 2002
 - Mental Arithmetic 3rd Class: Certified on February 2001

Skill Stack

- Languages
 - Japanese: Native Proficiency
 - English: Full professional Proficiency
- Development
 - **Backend: Ruby(Ruby on Rails), Python**
 - Frontend: TypeScript + React.js, TypeScript +Vue.js
 - **Database: MySQL**, PostgreSQL, MongoDB
 - **Architecture: Monolith, Modular Monolith**
 - Hosting: AWS ESK
 - Versioning: Git, GitHub
 - **CI/CD: GitHub Actions**, ArgoCD
 - Monitoring: Datadog, Sentry

Work Histories

Period	Genre	Position	Jobs
Aug 2021 - Present	SaaS(Educational Service)	Software Engineer	• Backend Development • Frontend Development • CI/CD Maintenance • Weekly Release Manager • Troubleshooting • Documentation • Technical Debt Resolution • Tech Blog Entries • LT Speaker
Feb 2020 - Dec 2020	SaaS(Chatbot Platform)	Backend Engineer	• Backend Development • Documentation • Technical Debt Resolution • Inspection of Alternative API

Work Histories

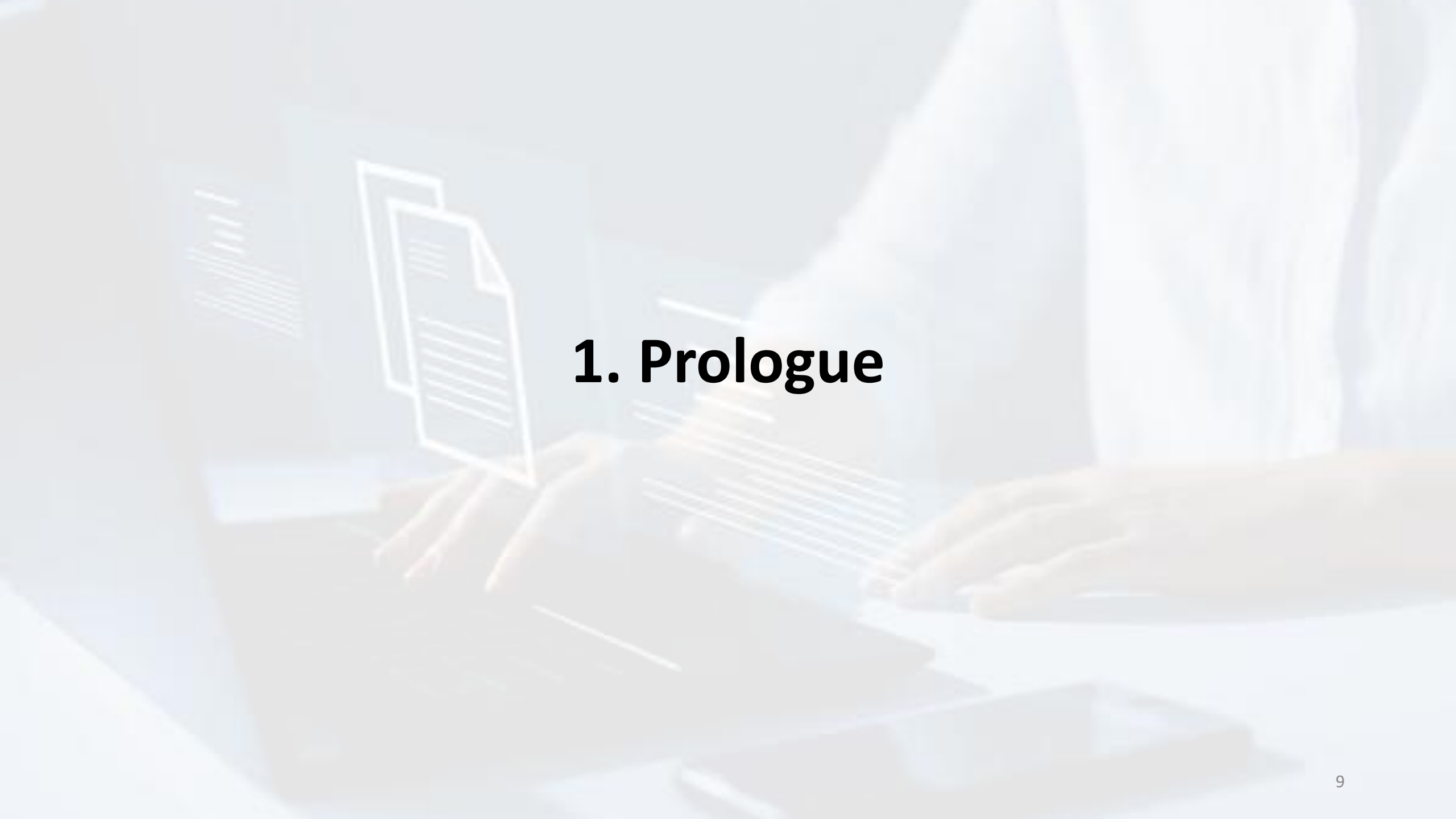
Period	Genre	Position	Jobs
Jun 2018 - Jan 2020	Contract-basis Developer	Software Engineer	• Backend Development • Frontend Development • Troubleshooting • Quality Assurance • Tech Blog Entries
Apr 2016 - Jan 2018	System Engineering Service	System Engineer	• Corporate Account Admin • Windows Server Maintenance • Security Admin Assistant • Internationalisation / Localisation

International-Exchange Activities

- Activities in University
 - English Linguistics Seminar(Focusing on Mass Media English)
 - International-Exchange Clubs(The 2nd Year)
 - [International-Exchange Programmes conducted by Japan Cabinet Office](#)(2013 - 2016)
 - Japanese Linguistics Course(The Final Year)
- Overseas Life Experience
 - Working Holiday in Australia(April 2014 - March 2015)
 - Language School for 1 month in Sydney
 - Work for 6 Months in [Hamilton Island Resort](#)
 - Volunteering for 1 Month as Assistant Teacher of Japanese Language at [St Ives High School](#) in NSW
- Other Activities
 - Keep Everyday Journal in English (April 2014 - Present)
 - [Sunrise Toastmasters Club](#)(February 2017 - March 2018)
 - [Vital Japan](#)(January 2018 - July 2019, October 2022 - February 2023)
 - Self Learning and Training of English Language
 - Participation Ruby-Related Tech Conferences

Agenda

1. [Prologue](#)
2. [What Documentation Is](#)
3. [Top 6 Required Factors for Corporate Technical Documentation](#)
4. [Top 5 Required Skills for Corporate Technical Documentation](#)
5. [Top 3 Benefits of Well-Organised Corporate Technical Documentation](#)
6. [Top 5 Inevitable Trade-Offs](#)
7. [Summary](#)
8. [References](#)

A person wearing a white lab coat is seated at a desk, typing on a laptop. The image is heavily blurred and has a light blue/white color overlay. Several white, stylized icons of documents or folders are floating in the air above the laptop keyboard, suggesting a digital or data-related context.

1. Prologue

1. Prologue

Corporate technical documentation one of the inevitable task which we must not avoid facing if we truly wish to smooth development and understanding requirements processes.

I myself have worked on documentation for a wide range of stakeholders, in which I designed, composed and released corporate technical documents for better project and team management.

I learned what domain knowledge the document must contain and how I should make complicated and profound descriptions understandable.

I have undergone loads of tough experiences and learned why, what, when, who and how we must do.

Now, it is highly likely that I should share what I learned and executed.

1. Prologue

However, just sharing my own knowledge and experience is definitely going to be highly subjective, arbitrary, and full of prejudice.

I relearned the principles of technical documentation for engineers and analysed what factors are essential for corporate technical documentation.

[*Basics Of Technical Documentation For Engineers*](#), Ayca LITTLE, Published in [*DatasheetEST by TDSmaker*](#), 25 April 2018

The article above focuses on the basics from the point of view of marketing, but they are applicable enough once I switch the viewpoint from marketing to engineering. That is why I was willing to adopt it as a reliable reference.

A person wearing a white lab coat is shown from the chest down, sitting at a desk and typing on a laptop. The image is faded and serves as a background. Overlaid on the left side of the image is a white line-art icon of a document with a folded corner. The text '2. What Documentation Is' is centered in a bold, black, sans-serif font.

2. What Documentation Is

2. What Documentation Is

The definition of documentation is **an act to communicate information via a document or the composed document itself.**

In other words, **documentation is one of the communication form.**

Basically, the communication is one-way is from the write to readers, so **the write is 100% responsible for the contents.**

Readers are stakeholders who are involved in a corporate organisation such as a team or project.

A corporate technical document is the most important interface between the writer and stakeholders, so how hard they contribute to the organisation heavily depends on the quality of the document.

A person wearing a white lab coat is shown from the chest down, sitting at a desk and typing on a laptop. The image is faded and serves as a background. A white outline of a document with lines of text is overlaid on the left side of the laptop screen. The text '3. Top 6 Required Factors for Corporate Technical Documentation' is centered in the middle of the image in a bold, black font.

3. Top 6 Required Factors for Corporate Technical Documentation

3. Top 6 Required Factors for Corporate Technical Documentation

To secure the quality of a corporate technical document as the most important interface between the writer and readers, the following 6 requirements must be fulfilled.

1. [To Be Scientific](#)
2. [To Be Efficient](#)
3. [To Be Impressive](#)
4. [To Clarify Who Readers Are](#)
5. [To Communicate between Developers and Non-Developers](#)
6. [To Design Appropriate Workflows](#)

3. Top 6 Required Factors for Corporate Technical Documentation

1. To Be Scientific

The following 3 elements are required to keep a corporate technical document scientific.

1. **Consistency:** Reproductive whenever, wherever and whoever
 - → Not influenced by any external factor
2. **Accuracy:** Accessible fact-based information
 - → All information is of supportive data or reference.
3. **Relevance:** Comprehensive provision of required information
 - → Particularly, the purpose of each task is a must item to be listed.

For the document to be scientific, **idempotence must allow readers to obtain all required fact-based information via the document and reproduce the same result at any time.**

3. Top 6 Required Factors for Corporate Technical Documentation

2. To Be Efficient

No matter how [scientific](#) a corporate technical document is, readers will absolutely take loads of pains at locating what they truly would like to access if required information is scattered.

It is one of the very writer's tasks to index and organise the document in order for readers to easily find the information they would like.

The writer is required to gather, index and organise all required information in the document, and then readers can access to what they truly reach for with the least efforts.

3. Top 6 Required Factors for Corporate Technical Documentation

3. To Be Impressive

No matter how scientific and efficient a corporate technical document is, it makes no sense unless readers bear in mind what the document mentions.

Unimpressive documents are often boring, uncomfortable to read and lacking in visual materials such as a sequence chart, a class chart and an ER(Entity-Relation) chart.

When we human beings vividly memorise something, our intrinsic emotions play an important role in a memory helper.

In addition, we tend to make a final decision based on our emotion, not on logics.

That is why **impressive forms, expressions, descriptions and visual information are highly helpful to appeal to catch readers' attention and interests.**

3. Top 6 Required Factors for Corporate Technical Documentation

4. To Clarify Who Readers Are

In reality, it is as impossible to create a corporate technical document which is understandable for all readers without exception as inventing a cuisine which satisfies all people around the world.

That is why **the write have to define who are the readers and what domain and technical prerequisites they are expected to fulfil in advance.**

- Developer only: Priority is **detailed** and **comprehensive technical** descriptions.
 - An agreement of what domain and technical knowledge are required is needed at least.
- Non-Developer included: Priority is **concise** and **understandable** descriptions
 - Technically details description can confuse non-developers.

3. Top 6 Required Factors for Corporate Technical Documentation

5. To Communicate between Developers and Non-Developers

It sounds cynical, but corporate technical documentation is not self-concluded within software engineers.

That is because **we make the most of technology in order to meet demands of the end users by fulfilling the requirements shared by someone on the business division.**

They make sense of what requirements and need must be met more precisely than developers because they are close to the end users.

The opposite is also equally true and developer understand technical restrictions which prevents the requirements from being fulfilled more accurately.

It is absolutely essential to establish close communication between developers and someone on the business division, which enables the write to create a high-quality corporate technical document which reflects the trade-offs mentioned above.

3. Top 6 Required Factors for Corporate Technical Documentation

6. To Design Appropriate Workflows

The writer must not share a corporate technical document whether it is public or private.

It is required to follow the well-organised workflow where design, composition, review and release of the document is handled by an authorised member within the agreed score.

- **Design:** Reader-friendly structure providing comfortable searching experiences
- **Composition:** Comprehensive provision of all required information with impressive descriptions
- **Review:** Feedback to the writer about something reader-unfriendly in terms of former elements
- **Scope:** Sharing information only with the specific stakeholders
- **Authorisation:** Assign a member who is in charge of all phases above
 - $\equiv$ Ownership. It should not belong to only a member but multiple ones in terms of load balancing and mid-to-long term management.

A blurred background image showing a person's hands typing on a laptop keyboard. A white outline of a document with lines of text is overlaid on the left side of the image.

4. Top 5 Required Skills for Corporate Technical Documentation

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To secure the quality of a corporate technical document as the most important interface between the writer and readers, the following 5 skills are required.

1. [Ability to Get Readers Well-Informed](#)
2. [Ability to Collaborate with Stakeholders](#)
3. [Ability to Solve a Problem by Oneself](#)
4. [Ability to Manage Resources and Costs](#)
5. [Ability to Predict Potential Issues](#)

4. Top 5 Required Skills for Corporate Technical Documentation

1. Ability to Get Readers Well-Informed

Technical and domain knowledge is often so abstract that the writer take loads of pains at put it into words, but it must be described as a concrete and imaginable object for readers.

It requires the writer to have an ability to understand enough what they are trying to verbalise and choose the best way to explain it.

In short, **the writer is required to be able to express something profound as something scientific, efficient and impressive as possible.**

4. Top 5 Required Skills for Corporate Technical Documentation

2. Ability to Collaborate with Stakeholders

It is quite crucial to keep a corporate technical document up to date based on [the communications between developers and non-developers](#).

Also, the active communications has an fantastic effect on maintenance and improvement of motivation and productivity among the stakeholders.

On the contrary, poor occasions of communication would degrade the quality of the document brought by poor engagement and creativity.

4. Top 5 Required Skills for Corporate Technical Documentation

3. Ability to Solve a Problem by Oneself

Although [the communications between developers and non-developers](#) is highly important, **developers are basically required to solve a problem by themselves in which process they have got to detect the root cause and show a solution to it.**

What is more, the independent competency enables them to make sure what information is lacking and how they should obtain via a specific communication path and vice versa.

If the writer fails to make sense of how they should understand required expertise, techniques, procedures and communication to put what they are trying to put into a technical document, they are highly likely not to have an enough ability to solve a problem by themselves.

4. Top 5 Required Skills for Corporate Technical Documentation

4. Ability to Manage Resources and Costs

In order to fulfil [the top 6 required factors for corporate technical documentation](#), the write is also necessarily required to be accurately sure and manage how much human and physical resources are needed to create the document and get rid of all trade-offs which can prevent it from being completed.

No matter how much the write is prepared, **they will definitely face much contingency** and find it difficult to make it as originally planned.

However, **the careful preparation is more meaningless than quite helpful to avoid making the same mistake by analyse what caused the gaps between the plan and result.**

4. Top 5 Required Skills for Corporate Technical Documentation

5. Ability to Predict Potential Issues

The objects of documentation are variable as time goes by because business is always going forward, so a corporate technical document is nothing but a snapshot of a certain moment.

That is why **it needs continuous updates**, but it takes a bunch of human and physical costs to reflect the difference every time something changes.

A smart measure is to contain such a potential change or have a placeholder(e.g., TBA/TBD) on the document, which will make it easier to put it into the documents once the details become crystal enough.

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5. Top 3 Benefits of Well-Organised Corporate Technical Documentation

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I explained SHOULDs so far, but what profits do you enjoy from a well-organised corporate technical document?

The answer is the following 3 benefits.

1. [Growth of Trust in Your Organisation](#)
2. [Mid-to-Long Term Advancement of Productivity](#)
3. [Mid-to-Long Term Cost Reduction](#)

5. Top 3 Benefits of Well-Organised Corporate Technical Documentation

1. Growth of Trust in Your Organisation

It is essential for a team or a project to integrate all required information in a [scientific](#), [efficient](#) and [impressive](#) way in order to handle onboarding and daily tasks.

Whether it has a corporate technical document kept up to date or not makes a huge difference in that a psychological burden would make weary bones of the stakeholders.

For example, imagine someone assigned to a team or a project where they have no idea what and how.

What is worse, their coworkers insist on that there is no document which shows the information they would like and they should make it done by yourself.

In reality, some software engineers are likely to be undergoing such a situation, and it is clearer than water that they are taking pains and terribly stressed out.

5. Top 3 Benefits of Well-Organised Corporate Technical Documentation

1. Growth of Trust in Your Organisation

In contrast, their psychological condition will be totally different should all required information be always available on the document and they be sure what they can fall back on even though they encounter something unknown.

What is better, how the document is maintained vividly reflects how the organisation is managed.

It means that **the stakeholders are consciously or unconsciously evaluate and rely on the organisation on the well-organised document basis.**

5. Top 3 Benefits of Well-Organised Corporate Technical Documentation

2. Mid-to-Long Term Advancement of Productivity

It seems that creating corporate technical documentation is costly in the short term. However, **it will absolutely return the investment in the mid-to-long term if the document allows a new comer to be quick in the uptake and proceed to advanced tasks, which also helps to improve their motivation and productivity.**

In the sense, **should your organisation have no such a document, it is wiser to prepare for it right now because it is the very initial cost you have to pay.**

On the contrary, someone new would keep strained on the onboarding phase and bring no benefit to your organisation were you to be reluctant invest the inevitable cost.

5. Top 3 Benefits of Well-Organised Corporate Technical Documentation

3. Mid-to-Long Term Cost Reduction

Once the writer have created and released a corporate technical document to the stakeholder, they are supposed to keep referring to it.

The document plays a role in a proxy which keeps providing the stakeholders with the corresponding answer to a specific question with idempotences.

As a result, no more costs are needed for each redundant query and heavy dependency on a specific member can be exterminated.

In this sense as well, it is wiser to prepare for it right now should your organisation have no such a document, because the document carries a huge profit to your organisation in the mid-to-long term.

A person wearing a white lab coat is shown from the chest down, sitting at a desk and typing on a laptop. The image is faded and has a soft, ethereal quality. Overlaid on the left side of the laptop is a stylized, white-outlined icon of a folder or document with lines representing text. The background is a light, neutral color.

6. Top 5 Inevitable Trade-Offs

6. Top 5 Inevitable Trade-Offs

I have explained *Science of Corporate Technical Documentation for Software Engineers -Theories-*. However, what I shared so far are just sheer idealism, as it were. Sadly, there are some blockers which hinder us from realising it. Corporate technical documentation has the following 5 trade-offs at least.

1. [Cynical Recursive Dependency of Work](#)
2. [Agreements to Make with Team Members](#)
3. [The More Stakeholders, The Less Quality](#)
4. [Responsibility of Continuous Maintenance](#)
5. [Difficulty in Find Out the Upper Limit of Investment](#)

How we should handle trade-offs above are supposed to be discussed the next deck, [*Science of Technical Documentation for Software Engineers -Practices-*](#).

A person wearing a white lab coat is seated at a desk, typing on a laptop. The laptop screen displays a stylized icon of a document with lines of text. The background is a soft, out-of-focus light blue.

7. Summary

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1. Top 6 Required Factors for Corporate Technical Documentation

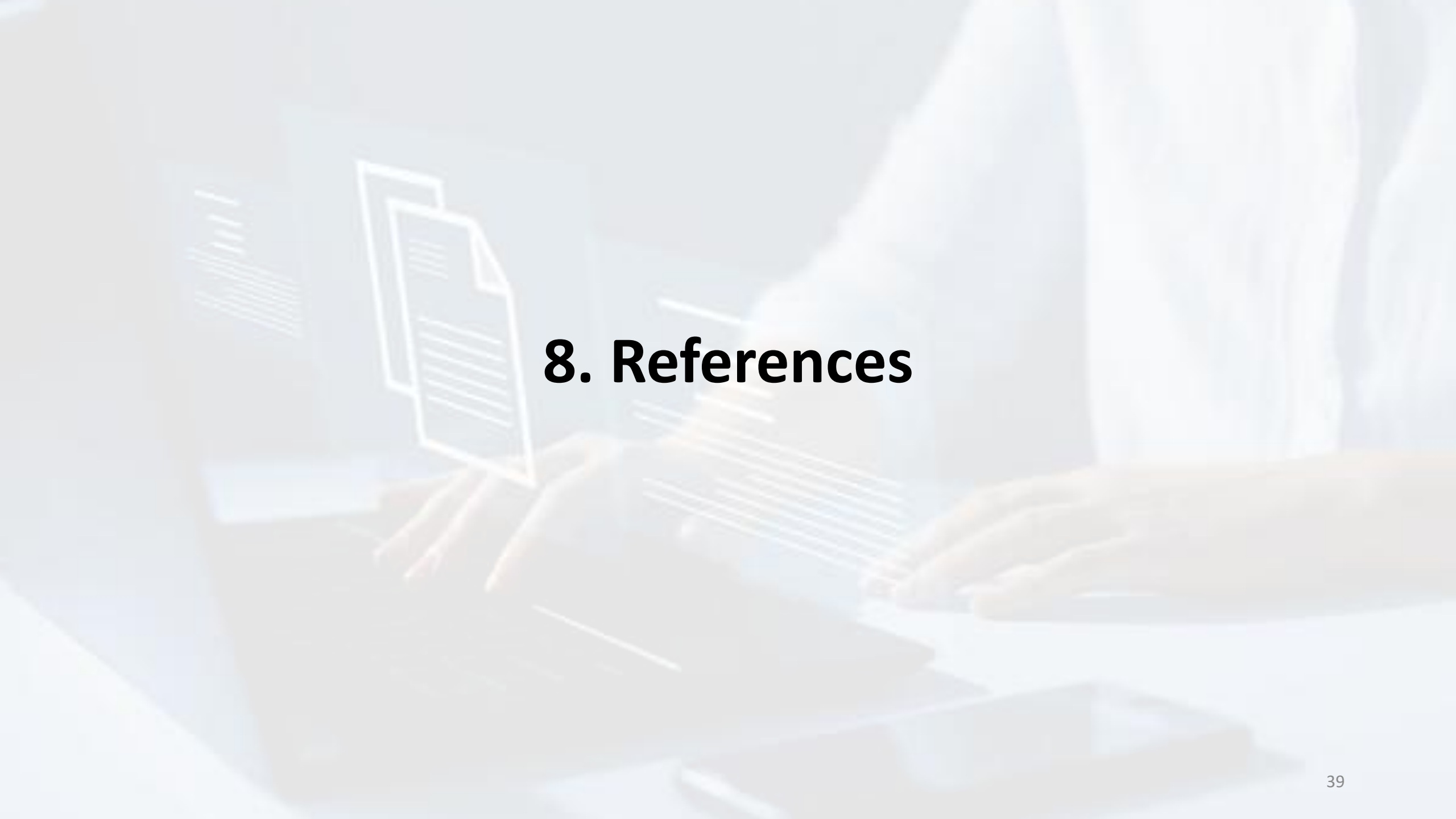
1. To Be Scientific
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3. To Be Impressive
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2. Top 5 Required Skills for Corporate Technical Documentation

1. Ability to Get Readers Well-Informed
2. Ability to Collaborate with Stakeholders
3. Ability to Solve a Problem by Oneself
4. Ability to Manage Resources and Costs
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3. Top 3 Benefits of Well-Organised Corporate Technical Documentation

1. Growth of Trust in Your Organisation
2. Mid-to-Long Term Advancement of Productivity
3. Mid-to-Long Term Cost Reduction

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8. References

8. References

- [Basics Of Technical Documentation For Engineers](#)
 - Last Accessed On: 12 November 2023
- [Dictionary.com](#)
 - Last Accessed On: 12 November 2023
- [英辞郎 Pro Lite](#)
 - Last Accessed On: 12 November 2023

A person wearing a white lab coat is shown from the chest down, sitting at a desk and typing on a laptop. The image is heavily blurred and has a light, ethereal quality. Overlaid on the scene are several white, glowing digital elements: a folder icon with a document inside, a series of horizontal lines, and a stylized representation of a document or data flow. The overall aesthetic is futuristic and technological.

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